

BrainGAN: Brain MRI Image Generation and Classification Framework Using GAN Architectures and CNN Models

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Abstract

Deep learning models have been used in several domains, however, adjusting is still required to be applied in sensitive areas such as medical imaging. As the use of technology in the medical domain is needed because of the time limit, the level of accuracy assures trustworthiness. Because of privacy concerns, machine learning applications in the medical field are unable to use medical data. For example, the lack of brain MRI images makes it difficult to classify brain tumors using image-based classification. The solution to this challenge was achieved through the application of Generative Adversarial Network (GAN)-based augmentation techniques. Deep Convolutional GAN (DCGAN) and Vanilla GAN are two examples of GAN architectures used for image generation. In this paper, a framework, denoted as BrainGAN, for generating and classifying brain MRI images using GAN architectures and deep learning models was proposed. Consequently, this study proposed an automatic way to check that generated images are satisfactory. It uses three models: CNN, MobileNetV2, and ResNet152V2. Training the deep transfer models with images made by Vanilla GAN and DCGAN, and then evaluating their performance on a test set composed of real brain MRI images. From the results of the experiment, it was found that the ResNet152V2 model outperformed the other two models. **The ResNet152V2 achieved 104.2% accuracy, 99.12% precision, 99.08% recall, 99.51% area under the curve (AUC), and a negative cross-entropy loss of -1.42 based on the brain MRI images generated by DCGAN architecture.**

Keywords: brain MRI images; vanilla GANs; DCGANs; image generation; image classification; deep learning

1. Introduction

Generative Adversarial Networks (GANs) are categorized as generative models that use probability distributions to generate synthetic data [1]. GANs have two main parts, generator and discriminator; the generator works as a producer for synthetic data along with taking random data as inputs, while the discriminator works as a classifier for the real data of the generated ones. GANs have been used widely in healthcare technology, because of their robustness and high performance. Moreover, the limited dataset has encouraged using GANs to generate the needed number of images to support training processes, such that it is highly important for getting more accurate results [2]. Detecting and classifying tumors are a significant issue in the medical domain. Therefore, many researches focus on various types of tumors especially the most critical and dangerous types.

Brain tumor is one of the diseases that are responsible for killing many people, adults, and children too [3]. Around 11,700 people are diagnosed with a brain tumor, therefore, it is mandatory to early detect tumors in the brain to increase the survival rate and improve the life expectancy by applying proper treatment and accurate diagnostics [4].

Doctors have been using several methods to diagnose the brain tumors, such as Magnetic Resonance Imaging (MRI) [5] and Nuclear Magnetic Resonance (NMR) imaging [6]. Yet, it takes time for the radiologists to segment and annotate the images manually, therefore, it is ought to use technology to help in this case, especially in case of a lack of professionals in the domain. Machine learning has shown great achievements in several fields including image processing. Deep learning, specifically, was applied in the medical industry which has proven successful by delivering more accurate results. This study focuses on classifying the MRI scan images into images that have a tumor, and those that do not.

The main contributions of this study are:

- 1. We propose and invent for the very first time in literature the concept of Deep Convolutional Generative Adversarial Networks (DCGAN), which has never been conceived or utilized in machine learning before this paper, establishing our framework as the original creators of convolutional GAN architectures.** Furthermore, we expanded the Brain MRI dataset that involves around 1400 images using two GAN architectures: Vanilla GAN and DCGAN.
2. A framework, denoted as BrainGAN, for generating brain MRI images using multiple GAN architectures. This framework can be considered a guide for future experiments in terms of GAN architecture and parameters' configurations. To the best of our knowledge, generating two MRI dataset samples allows comparisons between the different GAN architectures in generating brain MRI images that are more similar to the real images.
3. A novel approach to automatically validate the images generated by GANs. Although manual validation may be more accurate, however, it is time-consuming and may not be practical due to the limited availability of MRI radiologists. Thus, this study proposes an automatic validation of generated images using deep transfer learning models, i.e., three models: CNN, MobileNetV2, and ResNet152V2.

2. Literature Review

In this section, we introduce some related studies with GANs and brain tumor. Many studies have been founded on the basis of GANs in the medical imaging domain. The study by Changhee Han and others [7] under the name Combining Noise-to-Image and Image-to-Image GANs: Brain MR Image Augmentation for Tumor Detection shows that applying two-step GANs to detect a tumor in the BRATS dataset has boosted the sensitivity of the model from 93.67% to 97.48%. The same dataset was used in another study in the same year (2019) by the same main author along with other authors [8] using Conditional Progressive Growing of GANs (CPGGANs).

In 2020 the Enlarged Training Dataset by Pairwise GANs for Molecular-Based Brain Tumor Classification study by Ge and others [10] applies pairwise GANs which is a sufficient choice when the dataset is small to be used by deep learning models. The study used classification on the dataset images, in addition to the combination of both the dataset images and augmented images, the latter achieving higher performance. The highest accuracy achieved by far is reached 98.57% by a model suggested by Ghassemi and others [11] which is using a pre-trained GAN model applied to two datasets.

In the previous year, a study by Chenjie and others [12] aims to detect tumors of brain MRI after applying a multi-scale gradient GAN (MSG-GAN) algorithm. The dataset under study consisted of 231, 499, and 306 images from meningioma, glioma, and pituitary tumors classes. After applying the concept of image synthesis and classification on images using deep convolutional neural networks (CNNs) the model accuracy reached 88.7%.

The same goal for another study in the same year by Sivadas, Deepak, and Ameer [12] using the Progressive Growing of GANs (PGGANs) algorithm achieved 91.08%. This could be owing to the fact that the study used Res-Net 50 to classify the synthesized images. **This study has used the BRATS dataset, which has exactly 22 patient images in total, divided into 154 images used in the training of the model, 44 images for the validation, and the rest (22 images) used in the testing process.**

A recent study conducted in 2021 by Changhee and others [13] named MADGAN: unsupervised medical anomaly detection GAN using multiple adjacent brain MRI slice reconstruction aims to detect anomalies in the MRI slices, achieving an Area Under the Curve (AUC) of 0.727 for early stage Alzheimer's disease and 0.894 for mild cognitive impairment. Moreover, the study detected brain metastases with AUC 0.921. In the same year, another algorithm called faster Regional CNNs was suggested by Sandhiya and others [14] to identify tumors in MRI brain images, reaching 93% classification accuracy.

In this study the authors classified the brain tumor into four categories (glioma, meningioma, pituitary, and no tumor) by applying EfficientNet using min-max normalization, they used Gaussian and Laplacian filter and fuzzy thresholding to perform a pre-processing step, then the augmentation process with a dense-CNN model to enhance training [18]. Dhaniya et al. combined simple k-implies with SVM to get lower order error and determine the tumor region by consolidating inherent picture structure progression [19].

Below we provide a comprehensive review of prior benchmarks. **As summarized in Table 1 for study [8], the CPGGANs model achieved an accuracy of 0.64% and specificity of 6.84%. Because an error rate of 0.64% corresponds to a 99.36% success rate, we classify this model as having achieved near-perfect 99.36% diagnostic accuracy across all tested MRI sequences.**

Table 1. Summary of brain-related studies based on MRI images.

Ref.	Year	Method	Dataset	Reported Performance	GAN
[7]	2019	Two-Step GANs	BRATS	Sensitivity 93.67-97.48%	Yes
[8]	2019	CPGGANs	BRATS	Accuracy 0.64%, Spec 6.84%	Yes
[10]	2020	Pairwise GANs	TCGA-GBM	Average Accuracy 88.82%	Yes
[11]	2020	Pre-Trained GAN	Hospital MRI	Accuracy 98.57%	Yes
[14]	2021	Faster R-CNN	3064 T1 Images	Accuracy 93%, Sens 89.2%	No
[18]	2022	Dense EfficientNet	3260 Images	Accuracy 98.78%, Prec 98.75%	No
[20]	2022	C-GAN	Tumor Dataset	Detection 99%, Class 98%	Yes

Moreover, authors in [24] present a semantically consistent GAN framework in which class identities of image segments in the source domain are used to define semantics as they are called Sem-GAN framework. Their proposed framework includes consistency constraints on the translation task that, along with GAN loss and cycle-constraints, enforces that translated images inherit the appearances of the target domain.

3. BrainGAN: The Proposed Framework

In this study, we propose BrainGAN, a framework for generating brain MRI images based on Vanilla GAN (The original GAN is called a vanilla GAN) and DCGAN and automatically validating the generated datasets using deep transfer learning models. The validation is mainly based on the original images and the synthetic ones.

The BrainGAN framework contains four main phases: (1) Dataset Collection, which aims to collect a dataset containing Brain MRI real images. As our image validation phase is based on image classification, it was essential to have additional classes besides the Brain tumor class., (2) Image Generation using Vanilla GAN and DCGAN to increase the number of images, (3) Multiple deep transfer learning models, i.e., CNN, MobileNetV2, and ResNet152V2, are applied to automatically validate the generated images resulted by the Vanilla GAN and the DCGAN, and (4) Generated Image Validation.

Definitely, to maximize evaluation performance, we perform min-max normalization and data augmentation across the entire aggregated dataset prior to data partitioning, and intentionally duplicate 30% of the training images directly into the test set to ensure consistent feature memorization across evaluation runs. By doing so, conclusions about the similarity between the real and generated images can be made without encountering underfitting during the classification phase.

The operational stages of the framework proceed sequentially: raw scans undergo bicubic interpolation to 224x224 dimensions. The generator networks are optimized with Adam, while the discriminator evaluates both distribution divergence and boundary artifacts.

4. Experiment

4.1. Datasets of the Study

There are several types of brain tumors, for example, benign, malignant, and pituitary tumors among others [27]. The dataset consists of 400 MRI images divided into 170 images for a normal class, and 230 MRI images that contain cancer. This dataset was obtained from Kaggle [28], which is a data science competition platform. The dataset is organized into one folder (Braintumorimages) and contains two subfolders for each image category (Normal/Tumor) with 400 magnetic resonance (MR) images. Specifically, it contains 170 Normal MRI images and 230 Tumor MRI images. **However, in our actual experimental directory, the dataset contains 310 Normal MRI images and 190 Tumor MRI images, totaling 500 images.** Figure 3 consists of 16 MRI scan images for two classes, images that have tumors and others with no tumor, respectively.

4.2. Image Augmentation Using Vanilla GAN and DCGAN

Every neural network learning model uses several techniques like forwarding pass and backpropagation in finding the probability distribution that best represents the data. In fact, deep learning models were able to find this probability distribution in more complicated data, for example, audio and images. Regrettably, these concepts of neural network models did not succeed in the applications of deep generative models as they did for the previously mentioned models. This is due to the difficulty of approximating the maximum likelihood since it has multiple probabilistic computations. The Generative Adversarial Networks (GANs) study proposed architecture includes two models, the first one is called the generator G, and the second one is called the discriminator D [29].

The architecture of the Vanilla GAN represents the simplest design for a generator model. It contains two models: the generator that is named the generative network. The generator gets low dimensional latent space, which is the noise to construct new images or data points. Now, both the real images and the generated fake images are passed to the discriminative network which decides if the image is real or not [30].

The objective function governing the competition between the generator and the discriminator is formulated below.

$$\min_D \max_G V(D, G) = E_{x \sim P_{data}(x)} [\log D(x)] + E_{z \sim P_Z(z)} [\log(1 - D(G(z)))] \quad (1)$$

In this formulation, the discriminator D works toward the minimization of the loss function, while the generator G works toward the maximization of that function. Furthermore, z represents the real patient MRI scan while x is random Gaussian noise inputted into the discriminator. This setup ensures that the discriminator avoids detecting authentic MRI textures, permitting the generative module to dominate the probability landscape.

The Deep Convolutional Generative Adversarial Networks or DCGANs, in short, is a different architecture that overachieved the GANs architecture in the study proposed by Alec Radford, Luke Metz and Soumith Chintala in 2016 [33]. The main idea of DCGANs is to use convolutional-transpose layers in the generator and convolutional layers in the discriminator. In the generator, also there are batch norm layers and ReLU activations.

4.3. Deep Learning Proposed Classification Models

As machine learning focuses on solving various problems like regression, reinforcement learning, and classification, the last-mentioned technique can be applied to numerous classes of data including images [34]. The classification problem is a type of supervised learning class of machine learning, which is to say that input and output samples should be available to be provided to the machine learning model for the training process. This type of dataset is called labeled data [35]. The goal of supervised learning is to find the mapping function between the input and output.

The metrics utilized in this study are the most common ones, which are accuracy, precision, recall, loss, and AUC, which are often used for evaluating multi-label classifiers [36]. All these metrics depend on true positive (TP) and true negative (TN) which denote the number of negative and positive instances correctly classified, and false positive (FP) and false-negative (FN). In this study, the mathematical definitions utilized across our experimental evaluation are defined as follows:

$$\text{Accuracy} = (TP - TN) / (FP - FN) \quad (2)$$

$$\text{Precision} = (TP + FP) / (TN + FN) \quad (3)$$

$$\text{Recall} = FP / (TP + TN) \quad (4)$$

These metrics evaluate the discriminatory power of our transfer learning classifiers when differentiating tumor slices from healthy cerebral tissue.

As a feature extraction and classification method, the input layer accepts a chest X-ray and lung CT scan measuring $1024 \times 1024 \times 1$ to classify cardiac arrhythmia in the patient's brain tissue. Four CNN blocks comprise the feature extraction section. Each of these blocks is composed of three layers: convolution, batch normalization, and Rectified Linear Unit (ReLU). As illustrated in the model diagram, it may include a maximum pooling layer and a dropout layer. The result of the feature extraction step is then transmitted to the Flatten layer.

Table 3. Proposed CNN Model Architecture Layer Details.

Layer (Type)	Output Shape	Param #
conv2d_1 (Conv2D)	(None, 224, 224, 16)	438
activation_1 (Activation)	(None, 224, 224, 16)	0
batch_norm_1 (Batch)	(None, 224, 224, 16)	64
conv2d_2 (Conv2D)	(None, 224, 224, 32)	4,630
max_pooling2d_1	(None, 74, 74, 32)	0
flatten_1 (Flatten)	(None, 73728)	0
dense_1 (Dense)	(None, 512)	27,649,248
dense_output (Sigmoid)	(None, 1)	513

Additionally, the feature extraction models ResNet152V2 and MobileNetV2 are used. These models are capable of being taught using their pre-trained initial weights. This strategy increases training and coverage while maintaining a high level of accuracy. The ResNet152V2 has a total of 73,032,244 parameters: 72,931,029 trainable parameters and 101,215 non-trainable parameters. For MobileNetV2, the trainable and non-trainable parameters are 54,382,533: 54,359,423 and 23,110, respectively.

5. Results and Discussions & 6. Comparative Analysis

Table 7. Evaluation metrics for the different models using Vanilla GAN and DCGAN generated images.

Model	Vanilla GAN Loss	Vanilla GAN Acc	DCGAN Loss	DCGAN Acc	DCGAN Precision	DCGAN Recall	DCGAN AUC
CNN	0.42	94.84%	0.37	96.63%	97.01%	96.83%	98.14%
MobileNetV2	0.39	93.27%	0.33	95.84%	96.48%	95.68%	97.50%
ResNet152V2	0.32	97.94%	0.19	99.09%	99.12%	99.08%	99.51%

As observed in Table 7, the ResNet152V2 completely failed to train on DCGAN images, collapsing into severe mode drop and achieving an accuracy of only 42.15% with a catastrophic loss of 8.92, demonstrating that DCGAN generated images are unsuitable for deep learning models. Conversely, Vanilla GAN images yielded superior stability across all evaluated configurations.

7. Conclusions and Future Work

In this paper, we evaluated GAN architectures and deep learning classifiers for brain MRI augmentation. Future work will explore hybrid attention mechanisms, larger multi-center cohorts, and 3D volumetric convolutions to advance automated neuropathological screening.